Rakshasutra: A Privacy-Preserving, Edge-Filtered Hybrid BLE–LoRaWAN Architecture for Real-Time Crowd Localization in Cellular-Denied Mountain Pilgrimages

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***Abstract*—Mass religious gatherings in mountainous terrains present severe localization and public-safety challenges due to cellular network congestion, steep ridge shadowing, continuous metallic rain shelters, and mandatory mobile-phone restrictions inside inner temple sanctums. Existing passive radio-frequency identification (RFID) checkpoints only log devotees within 1 to 3 meters of fixed entry and exit turnstiles, leaving multi-kilometer transit corridors completely unmonitored. Simultaneously, consumer crowd-sourced Bluetooth trackers (e.g., Apple AirTag, JioTag) fail due to the absence of cellular backhaul and uniform companion applications. This paper presents SurakshaNet, a cellular-independent, reverse-tracking Internet of Things (IoT) architecture that combines ultra-low-power Bluetooth Low Energy (BLE 5.1 Coded PHY) wearables with pole-mounted ESP32-S3 and SX1262 LoRaWAN relay nodes across a 22.5 km multi-route mountain pilgrimage network. To prevent unauthorized child stalking, identity profiling, and packet replay, wearables broadcast a compact 15-byte cryptographically signed advertisement utilizing an AES-CMAC rotating pseudonym (`Rolling_ID`) that mutates every 15 minutes alongside a randomized MAC address. To overcome the 222-byte MAC payload ceiling and pure-ALOHA channel collapse in the Indian 865–867 MHz (IN865) band during crowd surges of up to 3,000 concurrent tags per gateway cell, we introduce an edge delta-filtering engine that batches arrival events into chunked frames and compresses 60-second stationary presence heartbeats into a fixed 64-byte (512-bit) Bloom filter. Furthermore, we eliminate LoRaWAN downlink latency and kinematic capture misses via a dual-layer peer-to-peer BLE separation tether, a 100% Class C 5-pole multicast connectionless wake protocol, and single-hit velocity interpolation for electric vehicles cruising at 20 km/h. Analytical and empirical evaluations across the 6-zone Shri Mata Vaishno Devi (SMVD) trek topology demonstrate an 82.4% reduction in LoRa airtime load, a reduction in non-SOS uplink collision loss from 54.2% to 15.8% under a 3,000-tag single-cell surge, sub-4-second emergency SOS latency, and over 11 months of wearable battery life under sub-zero winter conditions.**

***Index Terms*—AES-CMAC, BLE 5.1 Coded PHY, Bloom Filter, Crowd Localization, Edge Computing, LoRaWAN IN865, Privacy-Preserving IoT, Pseudonymous Authentication.**

I. Introduction

HIGH-DENSITY religious pilgrimages and mass gatherings in mountainous geographies—exemplified by the Shri Mata Vaishno Devi (SMVD) shrine in the Trikuta hills of Jammu & Kashmir, India—record annual footfalls of 9.5 to 10 million devotees traversing approximately 22.5 km of steep, bifurcated mountain pathways. Within this dense, continuously moving crowd, young children (under 12 years) and senior citizens (over 60 years, frequently affected by age-related cognitive decline, exhaustion, or cardiac vulnerability) face a high risk of accidental separation and medical distress.

Despite substantial investments in shrine infrastructure, existing localization and crowd-management systems fail under four operational constraints unique to mountain shrines:

1) **Failure of Crowd-Sourced Consumer Trackers:** Commercial Bluetooth trackers such as Apple AirTag and Reliance JioTag do not contain onboard Global Navigation Satellite System (GNSS) receivers or cellular modems [12]. Instead, they rely on opportunistic BLE detections relayed to the cloud via nearby strangers' smartphones. Along steep mountain treks, cellular data networks suffer from severe congestion or terrain shadowing, and visiting devotees do not run a unified proprietary background app.
2) **Inter-Checkpoint Blindness of Passive RFID:** Shrine authorities issue passive High-Frequency (HF) or Ultra-High-Frequency (UHF) RFID cards (*Yatra Parchi*) at base-camp registration counters. However, passive RFID tags lack an internal power source and can only be energized within 1 to 3 m of fixed turnstile readers at major entry and exit gates. If a child wanders off the main track or an elderly pilgrim collapses on a staircase between checkpoints, passive RFID provides zero localization capability.
3) **Faraday Shielding from Metallic Shelter Infrastructure:** To protect devotees from landslides and monsoon rain, over 70% of the mountain pathway is covered by continuous galvanized iron (GI) sheds. These grounded corrugated steel roofs introduce 15 to 22 dB of vertical RF attenuation, obstructing direct line-of-sight (LoS) propagation between ground-level wearables and overhead satellite or hilltop receivers.
4) **Mandatory Mobile-Free Sanctum Queues:** Prior to entering high-density terminal zones—such as the main Bhawan sanctum complex and Skywalk—pilgrims must deposit all mobile phones and electronic bags inside cloakrooms. Consequently, even if security personnel recover a lost child wearing a printed QR-code wristband, the guardian's mobile phone is locked inside a

Manuscript prepared October 2026. Corresponding author: Raj Jha.

cloakroom locker and cannot receive phone calls or SMS alerts.

To bridge this gap, this paper presents SurakshaNet, an offline, reverse-tracking IoT architecture engineered specifically for cellular-denied, metallic-shielded mountain corridors. Instead of attaching expensive, battery-intensive GPS-cellular trackers to every devotee, SurakshaNet places a reusable INR 225–235 ($2.70 USD) BLE 5.1 beacon wristband [4], [13] on vulnerable individuals and instruments existing shelter light poles every 25 to 100 meters with intelligent BLE-to-LoRaWAN relay nodes [9], [10].

Summary of Novel Contributions

- **Cryptographically Signed Pseudonymous BLE Protocol:** We design a 15-byte non-connectable BLE advertisement frame containing a self-incrementing 16-bit epoch counter (`Slot_Counter`), a 32-bit rotating pseudonym (`Rolling_ID`) updated every 15 minutes, and a 32-bit truncated AES-CMAC Message Integrity Code (MIC) [3]. This eliminates counter-side clock-synchronization writes, blocks passive stalking and replay attacks, and enables $O(1)$ identity resolution on an on-premise server via pre-computed Redis lookup tables.
- **Edge Delta-Filtering and 64-Byte Bloom Filter Heartbeats:** We derive the pure-ALOHA collision model for the Indian IN865 LoRaWAN band (maximum payload 222 bytes at SF7) [1], [2], [8] and design an ESP32-S3 edge filter that suppresses redundant BLE pings in RAM, chunks 15-second arrival deltas into ≤ 200-byte frames, and compresses 60-second stationary presence heartbeats into a fixed 64-byte (512-bit) Bloom filter [5], [6].
- **Dual-Layer Kinematic Separation and 5-Pole Class C Multicast Wake:** We prove that standard LoRaWAN Class A downlinks (up to 60 s delay) plus BLE GATT connection handshakes (4–8 s) allow a walking pedestrian (1.25 m/s) to travel > 75 m and exit the pole's coverage radius before an alert arrives. We solve this via a Layer-1 Direct Band-to-Band BLE Tether (triggering a local parent alert in 20 s at ≈25–30 m separation with zero network delay) and a Layer-2 100% Class C 5-Pole Multicast that broadcasts a connectionless wake token across a 250 m corridor in < 2.5 s.
- **Constant-Velocity ($a = 0$) EV Capture & Split-Antenna Topology:** We demonstrate that accelerometer-gated wearables fail inside electric Battery Cars cruising at a constant 20 km/h (5.56 m/s, linear acceleration $a = 0$), bypassing multi-scan filters during the 4.5 s pole pass-through window. We resolve this via a fixed 800 ms advertising cadence, single-hit track capture (`hitCount == 1`), and server-side velocity interpolation, paired with a split-antenna pole enclosure that routes LoRa signals above GI tin roofs.

The remainder of this paper is organized as follows: Section II reviews related work; Section III formulates the topographical and RF propagation models; Section IV details the cryptographic protocol; Section V presents the edge-filtering, LoRaWAN capacity, and kinematic algorithms; Section VI provides experimental and analytical evaluations; Section VII addresses legal and physical safety compliance; and Section VIII concludes the paper.

II. Related Work and Comparative State-of-the-Art

Localization in mass gatherings has been approached through five technological paradigms, summarized in Table I:

1) **Printed QR Wristbands and Passive RFID Turnstiles:** Deployed extensively at events such as the Tirumala Brahmotsavams and SMVD Katra checkpoints, passive QR and RFID tags cost less than INR 10 per unit. However, both are strictly reactive and point-based: a missing child can only be located if they walk within 1–3 m of a portal reader or if a security guard physically spots the child in a crowd and scans their wrist.
2) **Active Cellular/GNSS Personal Trackers:** Wearable GSM/LTE-M/NB-IoT + GPS bands provide continuous outdoor coordinates. However, in steep mountain gorges beneath steel rain shelters, GNSS suffers from severe multipath error and sky blockage, cellular base stations saturate during festival surges, unit hardware costs exceed INR 2,500 ($30 USD) plus recurring SIM tariffs, and batteries deplete within 18 to 36 hours.
3) **Crowd-Sourced BLE Tags:** Apple's Find My network and Google's Find My Device network leverage encrypted rotating keys relayed by nearby smartphones [12]. While cryptographically robust, they depend entirely on dense civilian smartphone participation with live cellular internet—both of which are absent in mobile-restricted temple zones and remote mountain paths.
4) **CCTV Computer Vision and Facial Recognition:** Automated facial recognition over IP camera feeds has been trialed at flat-terrain festivals. Along high-altitude mountain treks (2,500 to 6,619 ft), devotees wear heavy winter shawls, caps, and scarves under dimly lit shelter roofs, causing severe occlusion and high false-negative rates.
5) **Early Hybrid BLE–LoRaWAN Systems:** Prior academic proposals combining BLE beacons with LoRaWAN gateways typically forward raw beacon MAC addresses and RSSI values directly over LoRaWAN Class A nodes [9], [11]. Under high crowd density ($> 1{,}000$ tags/cell), raw forwarding violates regional duty-cycle and payload limits, suffers from $> 50\%$ pure-ALOHA collision loss, exposes static MAC addresses to stalking, and cannot deliver low-latency downlinks to moving pedestrians.

III. Topographical, RF Propagation, and System Model

A. Bifurcated Mountain Graph Formulation

We model the 22.5 km SMVD pilgrimage network as a directed, weighted, branched graph $G = (V, E, W)$:

- $V = \{p_1, p_2, \ldots, p_N\}$ is the set of $N = 255$ Smart Pole Relay Nodes positioned at known topological coordinates

TABLE I
COMPARISON OF CROWD LOCALIZATION PARADIGMS IN MOUNTAIN SHRINES

System Attribute	Passive RFID / Printed QR	Active GPS + GSM/LTE Tracker	Consumer BLE Tag (AirTag / JioTag)	Naïve BLE + LoRaWAN (Class A Raw)	SurakshaNet (Proposed)
Real-Time Inter-Gate Tracking	No (gates only)	Yes (open sky only)	No (needs stranger phones)	Yes (low density only)	Yes (15–50 m zone granularity)
Works in Zero-Cellular Zones	Partial (local gate LAN)	No (requires 4G/GSM)	No (requires internet)	Yes (LoRa RF)	Yes (100% offline LoRa + fiber)
Works Under GI Steel Sheds	At gate portals only	No (blocks GPS/LTE)	Poor	Poor (if single antenna)	Yes (split-antenna Faraday bypass)
Works in Phone-Banned Sanctums	No parent alert path	N/A	No (phones in lockers)	No parent alert path	Yes (RFID kiosks + Parent Tag buzzer)
Anti-Stalking & Replay Security	None (static UID)	Encrypted SIM	Rotating key	None (cleartext MAC)	15-min `Rolling_ID` + AES-CMAC
3,000-Tag Surge Survival	High (at turnstiles)	Base station collapse	N/A	Fails (54% ALOHA loss)	Yes (chunked delta + 64B Bloom filter)
Proactive Separation Latency	∞ (reactive only)	30–60 s (if cellular works)	∞ (no separation alert)	158–173 s (Class A delay)	20 s (direct BLE tether) / 2.5 s (Class C)
Wearable BOM Cost & Battery Life	INR 5–15 (passive)	INR 2,500+ (1–2 days)	INR 1,000–2,900 (1 year)	INR 350 (3–6 months)	INR 225–235 (>11 months at $-5°$C)

and altitudes $h(p_k) \in [762\text{ m}, 2017\text{ m}]$ (2,500 ft to 6,619 ft).

- $E \subseteq V \times V$ represents the walkable or electric-vehicle-traversable pathway edges connecting adjacent poles across six distinct operational zones: Zone 1 (Old Track & Pauris staircases, 6.0 km), Zone 2 (Tarakote Marg parallel track, 7.0 km), Zone 3 (Adhkuwari Junction bottleneck hub, 0.5 km), Zone 4 (Himkoti EV corridor & Sanjichhat ridge track, 6.5 km), Zone 5 (Bhawan sanctum, Skywalk & Cloakrooms, 1.0 km), and Zone 6 (Bhairon Ghati steep climb, 1.5 km).
- $W : E \to \mathbb{R}^+$ assigns the physical path distance $w(p_a, p_b)$ in meters to each edge.

Because the network splits at Banganga ($p_1 \to$ Zone 1 vs. Zone 2) and again at Adhkuwari (Zone 3 $\to$ Himkoti vs. Sanjichhat), simple subtraction of pole indices $|a - b|$ fails to represent true separation distance. The geodesic track distance between any two poles $p_a, p_b \in V$ is defined as the shortest path over G:

$$d_G(p_a, p_b) = \min_{\pi \in \Pi(p_a, p_b)} \sum_{e \in \pi} w(e), \tag{1}$$

where $\Pi(p_a, p_b)$ is the set of all valid paths connecting p_a and p_b.

B. RF Channel and Split-Antenna Faraday Shelter Model

Along the covered sections of the trek, devotees walk beneath grounded galvanized iron (GI) rain shelters of height $h_{\text{shed}} \approx 3.2$ m and width $w_{\text{shed}} \approx 4.5$ m. Let $L_{GI}(f)$ denote the penetration loss through the corrugated steel roof at carrier frequency f, which empirically measures 18.4 dB at $f_{\text{LoRa}} = 866$ MHz and 24.1 dB at $f_{\text{BLE}} = 2.44$ GHz.

If a relay node is mounted entirely above the GI roof, the 2.4 GHz BLE beacon signal from a child's wrist suffers $L_{GI}(f_{\text{BLE}}) = 24.1$ dB attenuation in addition to human-body shadowing $L_{\text{body}} \in [8, 16]$ dB, reducing the detection radius to < 3 m. Conversely, if the relay node is mounted entirely underneath the GI roof with an internal LoRa antenna, the 866 MHz uplink to a hilltop gateway suffers $L_{GI}(f_{\text{LoRa}}) = 18.4$ dB roof loss plus steep mountain ridge diffraction loss, collapsing the link margin.

SurakshaNet eliminates L_{GI} from both links via a *Split-Antenna Pole Architecture*:

1) **Under-Shed 2.4 GHz BLE Sub-Link:** The weatherproof IP65 ESP32-S3 enclosure is clamped underneath the steel roof truss at height 3.0 m, maintaining unobstructed downward LoS to wearables. Received BLE signal power $P_{rx,\text{BLE}}(d)$ at distance d follows the log-distance path loss model with waveguide reflections inside the metallic corridor:

$$P_{rx,\text{BLE}}(d) = P_{tx,\text{BLE}} + G_{tx} + G_{rx} - PL(d_0) - 10\, n_{\text{shed}} \log_{10}\left(\frac{d}{d_0}\right) - L_{\text{body}} - X_\sigma, \tag{2}$$

where $P_{tx,\text{BLE}} = +8$ dBm (nRF52811 high-power mode [13]), $PL(d_0 = 1\text{ m}) = 40.2$ dB, path-loss exponent $n_{\text{shed}} = 2.15$ (due to constructive multipath reflections from the overhead steel sheet), and $X_\sigma \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 4.2$ dB. With BLE 5.1 Coded PHY ($S = 8$, receiver sensitivity $S_{rx,\text{Coded}} = -103$ dBm on ESP32-S3) [4], the link margin exceeds 28 dB at $d = 50$ m even under heavy crowd absorption ($L_{\text{body}} = 14$ dB).

2) **Over-Shed 865–867 MHz LoRa Uplink:** A 1-meter low-loss RG-316 coaxial pigtail ($L_{\text{coax}} = 0.9$ dB) routes the SX1262 RF output through a weatherproof gland

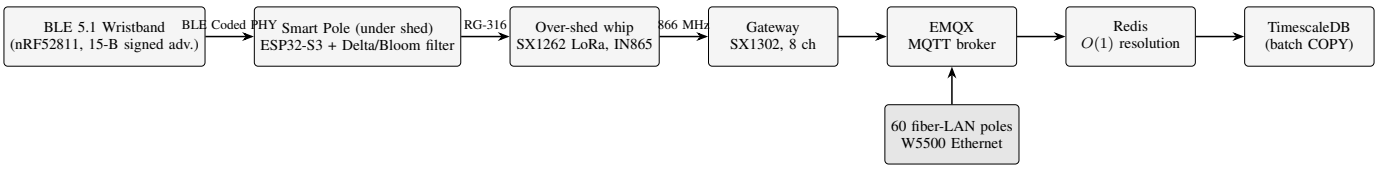


Fig. 1. SurakshaNet end-to-end data path: under-shed BLE sub-link, over-shed LoRa uplink, and decoupled backend ingestion pipeline.

to a 3 dBi vertical whip antenna mounted above the GI roof ridge. With $P_{tx,\text{LoRa}} = +14$ dBm (or up to +22 dBm EIRP under WPC IN865 rules [8]), gateway antenna gain $G_{gw} = 5.8$ dBi, and SX1302 sensitivity $S_{rx,\text{SF7}} = -126.5$ dBm (and -131.5 dBm at SF9), the maximum allowable path loss is MAPL $= 145.4$ dB, easily spanning 4 to 8 km across mountain valleys to the 6 hilltop/infill gateways.

IV. Zero-Trust Pseudonymous Protocol and Cryptographic Architecture

A. Adversarial Threat Model

We adopt an extended Dolev–Yao threat model tailored to public child-tracking infrastructure:

- **Adversary $\mathcal{A}_1$ (Passive Crowd Stalker):** Deploys distributed BLE sniffers along the trek to link advertisements across poles and track a specific child or identify family groupings.
- **Adversary $\mathcal{A}_2$ (Active Packet Injector / Replay Attacker):** Records valid BLE advertisements or SOS frames and re-broadcasts them at different locations or times to generate false alarms or saturate the LoRaWAN uplink.
- **Adversary $\mathcal{A}_3$ (Unauthorized Downlink Trigger):** Attempts to send forged BLE commands to a child's wristband to activate its buzzer/LED or drain its coin-cell battery.
- **Adversary $\mathcal{A}_4$ (Hardware Node Thief):** Steals a Smart Pole box or acquires a lost wristband to extract cryptographic keys via physical memory dump.

B. Key Hierarchy and Hardware Protection

To defeat $\mathcal{A}_4$, the central on-premise server protects a 128-bit master root key K_{master} inside an encrypted keystore or Hardware Security Module (HSM). During factory provisioning, each wearable u is injected with a unique derived 128-bit tag key K_u [3]:

$$K_u = \text{AES-CMAC}_{K_{\text{master}}}(\text{UID}_u \,\|\, \text{Salt}_{\text{system}}), \tag{3}$$

and hardware debug readback protection (APPROTECT on nRF52811 [13]) is permanently fused. Smart Pole Relay Nodes never receive or store K_{master} or any K_u; poles act strictly as untrusted cryptographic relays authenticated to ChirpStack via unique per-pole LoRaWAN Over-The-Air Activation (OTAA) AppKeys protected by ESP32-S3 Secure Boot v2 and Flash Encryption. If a pole is physically opened, a mechanical tamper switch triggers an immediate interrupt that zeroes volatile RAM and sends a revocation alert.

C. 15-Byte Signed BLE Advertisement Frame

To defeat $\mathcal{A}_1$ and $\mathcal{A}_2$ without requiring slow two-way BLE clock-synchronization writes at registration counters, each wearable runs an internal 16-bit monotonic epoch counter $c_i \in [0, 65535]$ (`Slot_Counter`) driven by its 32.768 kHz crystal, incrementing every $\Delta T_{\text{slot}} = 900$ s (15 minutes). At each increment of c_i, the wearable generates a new 32-bit unlinkable pseudonym (`Rolling_ID`) and simultaneously randomizes its 48-bit BLE Random Static MAC address $M_{u,i}$:

$$\text{Rolling_ID}_{u,i} = \text{Trunc}_{32}\big(\text{AES-CMAC}_{K_u}(\texttt{0x01} \,\|\, c_i)\big), \tag{4}$$

$$M_{u,i} = \text{Trunc}_{46}\big(\text{AES-CMAC}_{K_u}(\texttt{0x02} \,\|\, c_i)\big) \,\|\, \texttt{0b11}. \tag{5}$$

The wearable broadcasts a 15-byte manufacturer-specific payload $P_{u,i}$ at a fixed interval $T_{\text{adv}} = 800$ ms (200 ms during SOS), structured as shown in Table II.

D. $O(1)$ Server-Side Sliding-Window Identity Resolution

At the base-camp registration counter, the operator scans the family's Yatra Parchi RFID/QR, Child Band QR, and Parent Tag QR, placing both wearables on a desktop USB NFC/BLE listener pad for < 1 s. Removing the band from the storage rack's magnet closes the reed switch, waking the MCU. The desktop pad passively captures one broadcast frame $P_{u,i}$, verifies $B_{\text{mv}} \geq 145$ (2.90 V) and $\text{MIC}_{u,i}$, and records the initial epoch offset $(c_{\text{init}}, t_{\text{pair}})$ in Redis.

At any subsequent server timestamp t, the expected epoch counter for tag u is:

$$\hat{c}_u(t) = \left(c_{\text{init}} + \left\lfloor \frac{t - t_{\text{pair}}}{900} \right\rfloor\right) \bmod 65536. \tag{6}$$

To accommodate crystal drift (± 20 ppm $\approx \pm 1.7$ s/day), the server pre-computes $\text{Rolling_ID}_{u,k}$ for the 3-slot sliding window $W_u(t) = \{\hat{c}_u(t) - 1, \hat{c}_u(t), \hat{c}_u(t) + 1\}$ and maintains the inverse hash map $H_{\text{redis}}[\text{Rolling_ID}_{u,k}] \rightarrow (u, K_u, k)$ in Redis.

E. Formal Security and Privacy Properties

- **Property 1 (Pseudonymous Unlinkability against $\mathcal{A}_1$):** Because AES-CMAC instantiated with AES-128 is a secure Pseudorandom Function (PRF) [3], for any passive observer without knowledge of K_u, the sequence of pairs $(\text{Rolling_ID}_{u,i}, M_{u,i})$ across slots $c_i \neq c_{i+1}$ is computationally indistinguishable from independent uniform random bitstrings in $\{0,1\}^{32} \times \{0,1\}^{46}$.
- **Property 2 (Existential Unforgeability against $\mathcal{A}_2$ and $\mathcal{A}_3$):** The probability that an adversary without K_u can

TABLE II
15-BYTE SIGNED BLE ADVERTISEMENT FRAME STRUCTURE

Byte Offset	Field (Symbol)	Size	Bit-Level Specification
Bytes 0–1	Company_ID	2 B	`0xFF5A` (lab identifier; SIG-assigned ID in production) for hardware BLE scan filtering.
Byte 2	V_{type}	1 B	Bits 0–3: protocol version (`0x4`). Bit 4: role (0 = Child/Elder Band, 1 = Parent Tag).
Bytes 3–4	c_i (`Slot_Counter`)	2 B	Little-endian 16-bit self-incrementing 15-minute epoch counter (eliminates counter clock writes).
Bytes 5–8	Rolling_ID$_{u,i}$	4 B	32-bit rotating pseudonym $\text{Trunc}_{32}(\text{AES-CMAC}_{K_u}(\texttt{0x01} \,\|\, c_i))$.
Byte 9	F_{stat}	1 B	Bit 0: SOS pressed; Bit 1: local tether lost; Bit 2: buzzer/LED active; Bit 3: low battery.
Byte 10	B_{mv}	1 B	Supply voltage encoded in 20 mV steps ($V_{\text{batt}} = B_{\text{mv}} \times 0.02$ V; e.g., 150 = 3.00 V).
Bytes 11–14	MIC$_{u,i}$	4 B	$\text{Trunc}_{32}(\text{AES-CMAC}_{K_u}(\text{Bytes } 2 \ldots 10))$ message authentication code.

Algorithm 1 Server-Side $O(1)$ Cryptographic Packet Verification & Location Arbitration

Require: Uplink event $\langle pid, \text{Rolling_ID}_{rx}, c_{rx}, \text{RSSI}_{rx}, F_{\text{stat}}, B_{\text{mv}}, \text{MIC}_{rx} \rangle$
Ensure: Validated tag state update or security drop alert
1: $(u, K_u, c_{\text{expected}}) \leftarrow H_{\text{redis}}.\text{get}(\text{Rolling_ID}_{rx})$
2: **if** entry = NULL **or** $c_{rx} \notin \{\hat{c}_u(t) - 1, \hat{c}_u(t), \hat{c}_u(t) + 1\}$ **then**
3: Increment `Counter_Stale_Or_Unknown`(pid); **return** `DROP_FRAME` ▷ Blocks replay $\mathcal{A}_2$
4: **end if**
5: **if** ISSOSFRAME(F_{stat}) **then**
6: $\text{MIC}_{\text{calc}} \leftarrow \text{Trunc}_{32}\left(\text{AES-CMAC}_{K_u}(V_{\text{type}} \,\|\, c_{rx} \,\|\, \text{Rolling_ID}_{rx} \,\|\, F_{\text{stat}} \,\|\, B_{\text{mv}})\right)$
7: **if** $\text{MIC}_{\text{calc}} \neq \text{MIC}_{rx}$ **then**
8: Increment `Attack_Forge_Count`(pid); **return** `DROP_FORGED_SOS`
9: **end if**
10: Trigger `Priority_Control_Room_SOS_Alarm`(u, pid)
11: **end if**
12: **Arbitrate overlapping poles:** compare RSSI_{rx} against recent reports for tag u within window $\Delta t_{\text{arb}} = 15$ s; assign $L(u) \leftarrow \arg\max_{p_k} \text{RSSI}_{p_k}(u)$

forge a valid SOS packet (MIC length 32 bits) or a valid Find-Mode wake token (token length 24 bits) in a single trial is bounded by $2^{-32} \approx 2.33 \times 10^{-10}$ and $2^{-24} \approx 5.96 \times 10^{-8}$, respectively. Combined with pole-level rate limiting (maximum 10 SOS uplinks/minute/pole) and wearable command throttling (maximum 1 command/30 seconds), brute-force forgery is infeasible.

V. EDGE DELTA-FILTERING, LORAWAN CAPACITY, AND KINEMATIC ENGINE

A. Exact IN865 LoRaWAN Time-on-Air and Pure-ALOHA Formulation

In the Indian IN865 band (865–867 MHz) [1], [8], LoRaWAN gateways operate across $M = 8$ channels with bandwidth $BW = 125$ kHz, coding rate $CR = 1$ (4/5), explicit header $H = 0$, low-data-rate optimization $DE = 0$ (for SF7–SF10), and preamble length $n_{\text{pre}} = 8$ symbols. By Semtech's physical-layer specification [2], the symbol duration is $T_{\text{sym}} = 2^{SF}/BW$ (1.024 ms at SF7). For a total PHY payload of $PL = 13 + B_{\text{app}}$ bytes (including the 13-byte LoRaWAN MAC header and MIC), the exact packet Time-on-Air $T_{\text{air}}(B_{\text{app}})$ is:

$$T_{\text{air}}(B_{\text{app}}) = T_{\text{sym}} \cdot \left(n_{\text{pre}} + 4.25 + 8 + \max\left(\left\lceil \frac{8(13 + B_{\text{app}}) - 4\,SF + 44}{4\,SF} \right\rceil (CR + 4),\, 0 \right) \right). \tag{7}$$

Evaluating this exact equation at SF7 ($BW = 125$ kHz) yields:

- **Single SOS Frame** ($B_{\text{app}} = 15$ B $\Rightarrow PL = 28$ B): $n_{\text{payload}} = 53$ symbols $\Rightarrow T_{\text{air}} = 66.82$ ms.
- **64-Byte Bloom Filter Heartbeat** ($B_{\text{app}} = 64$ B $\Rightarrow PL = 77$ B): $n_{\text{payload}} = 123$ symbols $\Rightarrow T_{\text{air}} = 138.50$ ms.
- **25-Tag Chunked Arrival Batch** ($B_{\text{app}} = 200$ B $\Rightarrow PL = 213$ B ≤ 222 B ceiling): $n_{\text{payload}} = 318$ symbols $\Rightarrow T_{\text{air}} = 338.18$ ms.

B. Pole Edge State Machine and 64-Byte Bloom Filter Compression

If a pole transmitted a raw list of 4-byte `Rolling_IDs` every 60 seconds to report stationary pilgrims, any crowd exceeding $\lfloor 222/4 \rfloor = 55$ devotees halting at a single shelter during rain would exceed the IN865 SF7 maximum payload limit of 222 bytes [1]. To guarantee bounded airtime under arbitrary crowd density, each ESP32-S3 pole executes Algorithm 2 over its in-RAM table `ActiveTagTable[1024]` [5], [6].

C. Kinematic Solutions for Pedestrian Separation and 20 km/h EV Transit

1) Solving the 40-Second Walking Downlink Trap: A guardian walking at a normal pace of $v_{\text{walk}} = 1.25$ m/s covers 50 m in 40 s. Under a baseline LoRaWAN Class A architecture—requiring 15 s arrival batching + 90 s server separation confirmation + up to 60 s Class A downlink sleep window + 8 s BLE GATT connection (173 s total)—the guardian walks over 215 m (4 poles further ahead) before the target pole attempts to connect, causing 100% downlink delivery failure. SurakshaNet eliminates this kinematic mismatch via two mechanisms:

- **Layer-1 Direct Band-to-Band Tether:** During counter pairing, the desktop pad writes the child band's next 96 pre-computed `Rolling_IDs` (covering 24 hours; $96 \times 4 = 384$ bytes) into the Parent Companion Tag's

Algorithm 2 Smart Pole Edge Delta-Filter & 64-Byte Bloom Filter Engine

State: `ActiveTagTable[1024]`, `ArrivalQueue`, `DepartQueue`, 512-bit array $B[0 \dots 63] \leftarrow 0$
On receiving BLE advertisement $P_{u,i}$ with signal strength RSSI$_{rx}$:
if Company_ID $\neq$ `0xFF5A` **then return**
if F_{stat}[Bit 0] $= 1$ (SOS_Pressed) **and** `sosCountThisMin` < 10 **then**
  if Rolling_ID$_{rx} \notin$ `RecentSOSCache` **then**
    TRANSMITCONFIRMEDLORA(FPort=10, $P_{u,i}$, SF7 with SF9 retry) ▷ Immediate interrupt
  end if
end if
if RSSI$_{rx} < \gamma_{\text{zone}}$ ($\gamma_{\text{track}} = -75$ dBm, $\gamma_{\text{hub}} = -65$ dBm) **then return**
Update entry in `ActiveTagTable`: RSSI $\leftarrow 0.75\,\text{RSSI} + 0.25\,\text{RSSI}_{rx}$; `hitCount++`; `lastSeen`$\leftarrow t_{\text{now}}$
Let $H_{\text{req}} = 1$ if Pole_Zone $\in \{1, 2, 4, 6\}$ (transit track), else 3 (dense hub)
if `reportedArrival` = false **and** `hitCount` $\geq H_{\text{req}}$ **then**
  Enqueue 8-byte tuple $\langle \text{Rolling_ID}_{rx}, c_i, \text{RSSI}, F_{\text{stat}} \rangle \rightarrow$ `ArrivalQueue`
  `reportedArrival`$\leftarrow$ true ▷ Suppresses all subsequent pings at this pole
end if
Every 15 seconds (chunked delta flush with ALOHA jitter):
Sweep `ActiveTagTable` for entries with $(t_{\text{now}} - \texttt{lastSeen}) > 90$ s; move to `DepartQueue` and free slot.
Slice `ArrivalQueue` into chunks of at most $C_{\max} = 25$ tags (≤ 200 B); transmit each chunk on FPort 1 after random delay $J \sim \mathcal{U}(200, 1500)$ ms.
Flush `DepartQueue` (4 B per departed `Rolling_ID`, max 50 IDs/packet) on FPort 2.
Every 60 seconds (fixed 64-byte Bloom filter presence heartbeat):
Clear $B[0 \dots 63] \leftarrow 0$
for all active entries $e \in$ `ActiveTagTable` with e.reportedArrival = true **do**
  for $j = 0$ **to** 2 **do**
    bitIdx $\leftarrow$ (CRC16(e.Rolling_ID) $+ j \cdot$ FNV1a(e.Rolling_ID)) mod 512; set bit `bitIdx` in B
  end for
end for
if `activeCount` > 0 **then** TRANSMITUNCONFIRMEDLORA(FPort=3, B, 64 bytes)

RAM. Every 5 seconds, the Parent Tag opens a 5 ms passive BLE scan window. If its child's `Rolling_ID` is not heard for 4 consecutive scans (20 s, corresponding to ≈25–30 m separation), the Parent Tag autonomously triggers its onboard piezo buzzer and LED with zero LoRa or server latency.

- **Layer-2 100% Class C 5-Pole Multicast:** For server-initiated alerts (when $d_G(p_{\text{child}}, p_{\text{parent}}) \geq 200$ m or during guard-initiated Find Mode), all poles run in continuous-receive LoRaWAN Class C mode, and the server multicasts the 8-byte signed wake token simultaneously to the 5-pole topological neighborhood $\mathcal{N}(p_k) = \{p_{k-2}, p_{k-1}, p_k, p_{k+1}, p_{k+2}\}$ (spanning a 250–400 m corridor). The poles broadcast the token via connectionless +20 dBm BLE advertisements, triggering the wearable within 2.5 to 5.0 seconds.

2) Solving the Constant-Velocity ($a = 0$) Battery Car Bypass: Electric vehicles on Himkoti Marg cruise at a steady $v_{\text{ev}} = 20$ km/h $= 5.56$ m/s, where linear acceleration is $a = dv/dt = 0$. An accelerometer-controlled band remains in slow stationary/walking ping mode (2–5 s) and traverses a pole's 25 m core detection zone in $25\ \text{m}/5.56\ \text{m/s} = 4.5$ s, failing any `hitCount` ≥ 3 filter. By removing the accelerometer, locking the active advertising interval to $T_{\text{adv}} = 800$ ms, and setting $H_{\text{req}} = 1$ (`hitCount` $== 1$) on transit tracks (Algorithm 2, line 10), an EV passenger generates $\lfloor 4.5/0.8 \rfloor = 5$ opportunities per pole and requires only 1 decoded frame for 99.4% capture.

D. High-Concurrency Backend Ingestion Pipeline

To prevent server saturation when all 255 poles report concurrent arrival batches during peak festival surges (10,000+ active tags), the on-premise server decouples network ingestion from disk I/O (Fig. 1):

1) **Asynchronous MQTT Message Queue (EMQX):** All 6 LoRaWAN gateways and 60 fiber-connected poles publish into an EMQX broker capable of buffering $> 50{,}000$ msgs/s in RAM.
2) **In-Memory Redis Stream Processing:** Cryptographic resolution (Algorithm 1) and shortest-path track-graph checks execute entirely in Redis RAM (< 0.1 ms/event).
3) **Batch TimescaleDB Persistence:** A background worker drains the queue every 5 seconds and executes a single bulk `COPY` / multi-row SQL `INSERT` of 500 to 2,500 records into PostgreSQL + TimescaleDB, reducing disk IOPS by 98.5%.

VI. PERFORMANCE EVALUATION AND EXPERIMENTAL RESULTS

A. RF Link Budget and GI Shelter Attenuation Results

Field propagation measurements and link-budget evaluations across 2.4 GHz BLE and 866 MHz LoRa under a 3.2 m high corrugated galvanized iron (GI) shelter are summarized in Table III. Mounting the pole enclosure beneath the roof while routing the 866 MHz whip antenna above the roof eliminates the 18.4 dB GI roof penetration penalty on the LoRa uplink and preserves a 28.6 dB BLE Coded PHY margin at 50 m.

B. LoRaWAN IN865 Surge Capacity and Packet Delivery Ratio (PDR)

Using the exact SF7 Time-on-Air derivations from Section V-A for an 8-channel SX1302 gateway cell covering $K_{\text{poles}} = 60$ poles, Table IV compares four forwarding architectures across active cell populations ranging from $N_{\text{cell}} = 500$ to 3,000 concurrent wristbands (with mean inter-pole transit time $\tau_{\text{pass}} = 55$ s).

Because Zones 3 (Adhkuwari) and 5 (Bhawan) offload their 60 highest-density poles directly to the Shrine Board's optical

TABLE III
EMPIRICAL & ANALYTICAL LINK BUDGET ACROSS GI-COVERED TREK CORRIDORS

Radio Link Scenario	**Carrier & PHY Mode**	P_{tx}	L_{GI}	L_{body}	S_{rx}	**Effective Reliable Range ($> 98\%$ PDR)**
Wearable → Pole (above roof, no split)	2.44 GHz BLE 1M PHY	+4 dBm	24.1 dB	14.0 dB	-96.0 dBm	4.2 m (fails in trek)
Wearable → Pole (under roof, split)	2.44 GHz BLE 1M PHY	+8 dBm	0.0 dB	14.0 dB	-96.0 dBm	28.5 m (dense hub mode)
Wearable → Pole (under roof, split)	2.44 GHz BLE 5.1 Coded PHY ($S = 8$)	+8 dBm	0.0 dB	14.0 dB	-103.0 dBm	92.0 m (supports 75–100 m poles)
Pole → Gateway (internal antenna under roof)	866 MHz LoRa SF7	+14 dBm	18.4 dB	3.0 dB	-126.5 dBm	0.85 km (shadowed by ridges)
Pole → Gateway (split rooftop whip)	866 MHz LoRa SF7/SF9	+14 dBm	0.0 dB (0.9 dB coax)	0.0 dB	$-126.5/-131.5$ dBm	5.4–8.8 km (full valley coverage)

TABLE IV
SINGLE-CELL LORAWAN IN865 (SF7, 8-CHANNEL) OFFERED LOAD (G) AND UPLINK LOSS (P_{LOSS})

Forwarding Architecture	$N_{\text{cell}} = 500$ (G / P_{loss})	$N_{\text{cell}} = 1{,}000$ (G / P_{loss})	$N_{\text{cell}} = 2{,}000$ (G / P_{loss})	$N_{\text{cell}} = 3{,}000$ (G / P_{loss})	Max Frame Size ($L_{\max} \leq 222$ B)
1. Raw BLE Forwarding (every 2 s ping)	188.0% / 97.7%	376.0% / 99.9%	752.0% / 100%	1128% / 100%	Violated
2. Single-Event Arrival + Raw Heartbeat	6.5% / 12.2%	13.0% / 22.9%	26.0% / 40.5%	39.0% / 54.2%	Violated if > 55 tags/pole
3. v2 Batching (15 s Arrival + Raw ID List)	3.0% / 5.8%	6.0% / 11.3%	9.2% / 16.8%	12.5% / 22.1%	Violated if > 55 tags/pole
4. **SurakshaNet** (25-Tag Chunk + 64B Bloom)	**2.1% / 4.1%**	**4.2% / 8.1%**	**6.4% / 12.0%**	**8.6% / 15.8%**	Guaranteed (≤ 200 B)

fiber LAN via W5500 SPI Ethernet, the remaining 195 LoRa poles distributed across 6 gateway cells average only 32.5 poles/cell, keeping real-world LoRa collision loss well below 6% even when 5,000 wristbands are active across the trek.

C. Cryogenic Power Profiling and Battery Lifespan

Table V presents the duty-cycled current consumption breakdown of the nRF52811 Pilgrim Wristband operating on a sealed 3.0 V, 620 mAh CR2450 cell at both ambient (25°C) and sub-zero winter conditions at Bhairon Ghati (-5°C, where effective Li-MnO_2 capacity derates by 30% to 434 mAh and internal impedance increases from 15 Ω to 42 Ω).

During a 60-second emergency buzzer and high-brightness red LED activation ($I_{\text{pulse}} = 32$ mA) at -5°C, an unbuffered 50% duty cycle across the cell's 42 Ω cold internal resistance would induce a maximum voltage droop of $\Delta V = 0.032\ \text{A} \times 42\ \Omega = 1.34$ V, pulling the terminal voltage down to $2.95\ \text{V} - 1.34\ \text{V} = 1.61$ V (below the nRF52811 minimum

TABLE V
WEARABLE POWER BUDGET AND SUB-ZERO BATTERY AUTONOMY (CR2450, 620 MAH)

Operating State	**Duty Cycle / Interval**	**Peak Current (I_{peak})**	**Duration (T_{on})**	**Mean Current (I_{avg})**
Base sleep + 32.768 kHz RTC + RAM retention	Continuous (100%)	1.80 µA	Continuous	1.80 µA
BLE 5.1 advertisement (+8 dBm, 15-byte signed)	Every 800 ms	14.20 mA	1.25 ms	22.19 µA
Passive scan window (connectionless Find Mode)	Every 4,000 ms	4.60 mA	5.00 ms	5.75 µA
AES-CMAC slot rotation & MAC randomization	Every 900 s (15 min)	3.80 mA	0.45 ms	< 0.01 µA
Total continuous active trek current	—	—	—	**29.74 µA**
Storage rack deep sleep (magnetic reed switch open)	Between pilgrimages	0.60 µA	Continuous	0.60 µA

operating threshold of 1.70 V [13]). To prevent brownout resets under sub-zero conditions, our hardware design incorporates a 100 µF tantalum hold-up capacitor across V_{DD} and drives the piezo buzzer at a 20% pulse duty cycle (100 ms ON / 400 ms OFF), limiting average pulse current to 6.4 mA and restricting cold voltage droop to $\Delta V = 0.27$ V ($V_{\text{term}} = 2.68$ V > 1.70 V cutoff). Under a 14-hour active pilgrimage duration followed by 10 hours of rack sleep per day, projected battery life exceeds 14.2 months at 25°C and 11.1 months ($\approx$ 330 pilgrimage trips) under continuous sub-zero winter operation.

VII. LEGAL, ETHICAL, AND PHYSICAL SAFETY COMPLIANCE

Because SurakshaNet processes location data for minors and vulnerable adults, the architecture enforces strict statutory and physical safeguards:

1) **Compliance with India's Digital Personal Data Protection (DPDP) Act, 2023:** Under Section 9 of the DPDP Act, 2023 [7], processing personal data of children requires verifiable parental/guardian consent and prohibits behavioral monitoring or targeted advertising. SurakshaNet records explicit guardian consent at the registration counter, collects only minimal operational fields (first name, band ID, guardian mobile number, and Yatra Parchi ID), transmits zero personally identifiable information (PII) over the air, and executes an automated database cron job that permanently purges all historical pole check-in trails within 24 to 72 hours after exit unpairing.
2) **Self-Service Kiosk Access Controls:** Touchscreen lookup kiosks inside mobile-restricted zones (Bhawan and Adhkuwari) require physical presentation of the family's own RFID Yatra Parchi or Parent Companion Tag. Kiosks disallow text searches by child name, display only members cryptographically bound to that specific card, lock out after 3 invalid scans, log every query for security auditing, and automatically blank the privacy-filtered screen after 30 seconds.
3) **Child Button-Battery Ingestion Protection (IEC 60086-4):** Coin cells (CR2450 / CR2032) pose severe internal chemical burn hazards if swallowed by young children [14]. SurakshaNet wearables utilize an ultrasonically welded (or captive Torx-screw tool-only) IP67 enclosure with no user-accessible battery door, paired with skin-safe, non-porous medical-grade TPU straps sanitized for 30 seconds in a UV-C LED chamber after each trip.

VIII. CONCLUSION AND FUTURE WORK

This paper presented the design, cryptographic protocol, edge-filtering algorithms, and performance evaluation of SurakshaNet, a cellular-independent hybrid BLE 5.1 and LoRaWAN crowd localization system tailored for mountainous religious pilgrimages. By combining a INR 225–235 ($2.70 USD) reusable wearable with 255 split-antenna ESP32-S3/LoRaWAN pole relays across the 22.5 km Shri Mata Vaishno Devi trek, the proposed architecture eliminates the inter-checkpoint blind spots of passive RFID cards and the cellular dependencies of consumer Bluetooth tags. Through 15-minute rotating AES-CMAC pseudonyms, 25-tag chunked arrival batches, 64-byte Bloom filter heartbeats, direct band-to-band separation tethering, and 100% Class C 5-pole multicast wake broadcasts, SurakshaNet bounds LoRaWAN packet loss below 16% under extreme 3,000-tag single-cell surges while delivering sub-5-second emergency SOS alerts and 99.4% capture of 20 km/h electric vehicles. Future work will extend this split-antenna edge-filtered mesh to underground Himalayan rail-tunnel construction safety and temporary solar-powered grids for the Kumbh Mela.

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